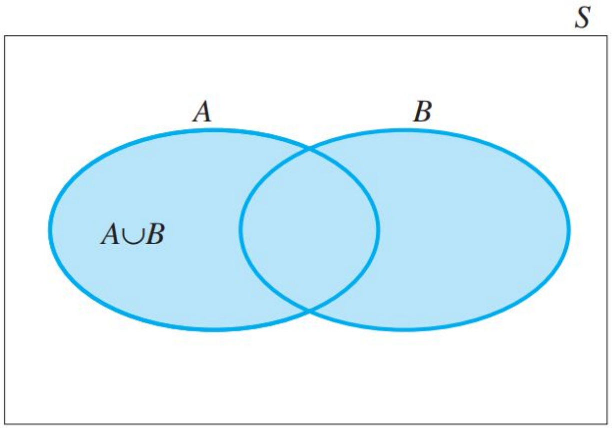
# Event Relations and Probability Rules

# Event Relation

Sometimes the event of interest can be formed as a combination of several other events. Need to consider 3 important event rules (same as set theory)

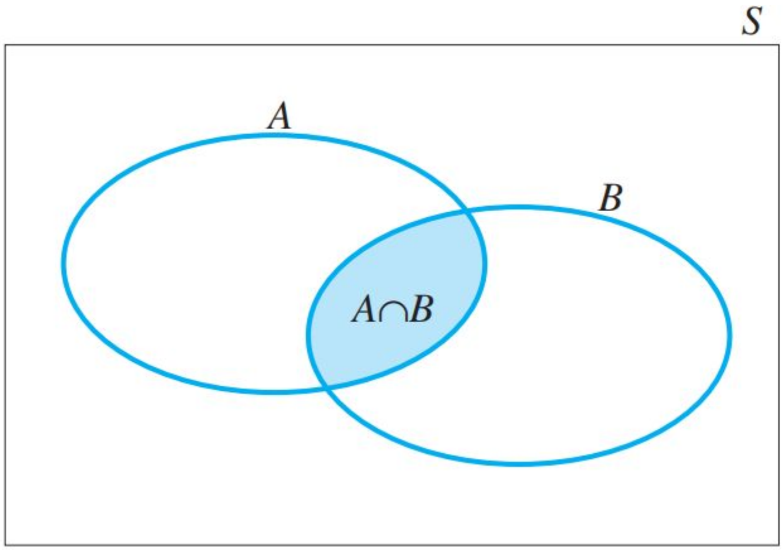
# The Union Rule

The **union** of events A and B, denoted by **A ∪ B,** is the event that either A or B or both occur.



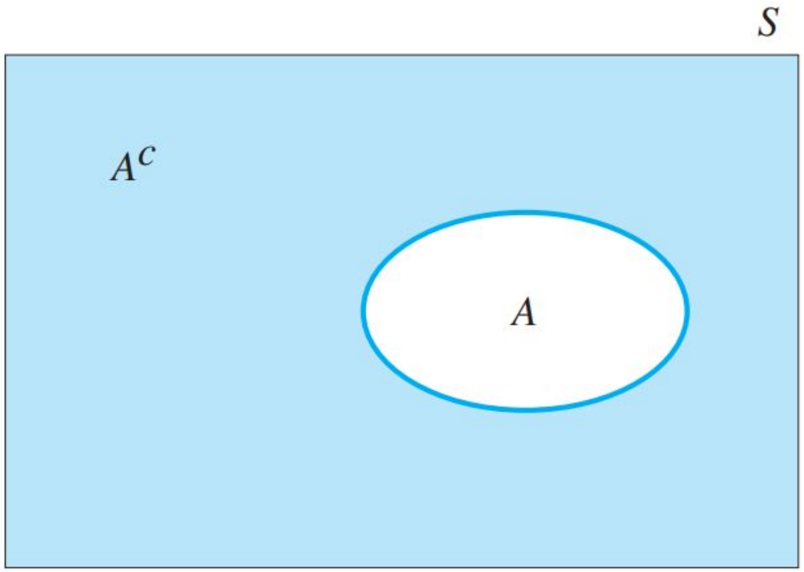
# Intersection Rule

The **intersection** of events A and B, denoted by **A ∩ B**, is the event that both A and B occur.



# Complement Rule

The **complement** of an event A, denoted by $A^c$ (or A'), is the event that A does not occur



# Example

Two fair coins are tossed, and the outcome is recorded. These are the events of interest:

A: Observe at least one head

B: Observe at least one tail

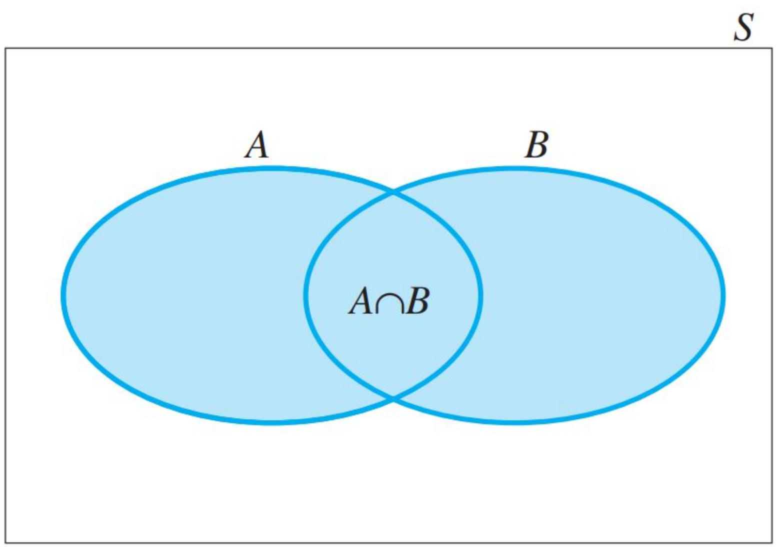
Define the events **A**, **B**, **A** ∪ **B**, **A** ∩ **B**, and $\mathbf{A}^c$ as collections of simple events, and find their probabilities

# Calculating Probabilities for Unions and Complements

# The Addition Rule

Given two events, A and B, the probability of their union, **A** ∪ **B**, is equal to

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



# Mutually Exclusive

What happens if two events are mutually exclusive?

# Rule for Complement

$$P(A) + P(A^c) = 1$$

Or

$$P(A) = 1 - P(A^c)$$

# Example

An oil-prospecting firm plans to drill two exploratory wells. Past evidence is used to assess the possible outcomes listed in Table

**Outcomes for Oil-Drilling Experiment**

| Event | Description | Probability |
|---|---|---|
| $A$ | Neither well produces oil or gas | .80 |
| $B$ | Exactly one well produces oil or gas | .18 |
| $C$ | Both wells produce oil or gas | .02 |

Find $P(A \cup B)$ and $P(B \cup C)$

The event $A \cup B$ can be described as the event that ***at most*** one well produces oil or gas, and $B \cup C$ describes the event that ***at least*** one well produces gas or oil.

# Independence, Conditional Probability and The Multiplication Rule

# Independent and Dependent Events

There is a probability rule that can be used to calculate the probability of the intersection of **several events**. However, this rule depends on the important statistical concept of independent or dependent events.

# Definition

Two events, A and B, are said to be **independent** if and only if the probability of event B is not influenced or changed by the occurrence of event A, or vice versa.

# Given

The probability of an event A, given that the event B has occurred, is called the conditional probability of A, given that B has occurred, denoted by $P(A|B)$

# General Multiplicative Rule

The probability that both A and B occur when the experiment is performed is

$P(A \cap B) = P(A) * P(B|A)$

## Conditional Probability

$P(A|B) = P(A \cap B) / P(B)$ Given $P(B) \neq 0$

$P(B|A) = P(A \cap B) / P(A)$ Given $P(A) \neq 0$